Shoaib Hassan

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AI/ML Engineer | Machine Learning | NLP Foundations | Python | SQL
Junior AI/ML Engineer with strong foundations in Python, machine learning, and SQL. Built an end-to-end machine learning pipeline on Karachi crime data (2020–2025), covering data preprocessing, feature engineering, model development, and evaluation. Developed a Random Forest classifier achieving ~86% accuracy and applied K-Means clustering to identify high-crime hotspots. Performed model evaluation using Accuracy, Precision, Recall, and F1-Score, then designed and implemented a star schema data warehouse in PostgreSQL to support multidimensional analysis and SQL-based reporting.

Bachelor of Science in Computer Science
Iqra University — Karachi, Pakistan
Graduation: Feb 2026
CGPA: 3.16

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL
- **Data Analysis & Visualization:** NumPy, Pandas, Matplotlib
- **Machine Learning:** Scikit-learn, K-Means Clustering, Random Forest, Classification, Feature Engineering, Model Evaluation (Accuracy, Precision, Recall, F1-Score), Confusion Matrix
- **Data Warehousing & Databases:** PostgreSQL (Supabase), Star Schema Design, Fact & Dimension Tables, SQL Aggregations, SQL Views
- **Tools & Platforms:** Git, GitHub, Jupyter Notebook, VS Code

PROJECTS

Crime Data Analysis & Hotspot Detection | (End-to-End ML + Data Warehouse)
https://github.com/shoaibhassancs/crime-data-analysis-and-hotspot-detection

- Built an end-to-end machine learning pipeline to analyze crime data from Karachi (2020–2025).
- Implemented K-Means clustering using latitude and longitude to identify high-crime hotspots and rank zones by intensity.
- Developed a Random Forest classification model to predict crime severity (High / Medium / Low) achieving ~86% accuracy.
- Engineered temporal and spatial features including peak hours, weekend indicators, and priority ranks.
- Designed and implemented a star schema data warehouse (Fact + Dimension tables) in PostgreSQL for multidimensional crime analysis.
- Created SQL views and aggregations to analyze monthly trends, high-severity areas/districts, and peak vs non-peak crime patterns.
- Performed feature importance analysis to identify key drivers of crime severity.

CERTIFICATIONS & COURSES

- **Google – Introduction to Generative AI:** Foundation models, Transformer architecture, Modern GenAI systems
- **Scikit-Learn Crash Course (Machine Learning with Python):** Supervised learning fundamentals, Model training and evaluation workflows
- **Python Fundamentals Series:** Core Python programming, Data handling with Pandas, Visualization with Matplotlib